A simple limiting regime for studying generalization analytically

Wide neural network in the NTK regime ≈ Kernel regression

To better understand generalization, one useful approach is to consider a limiting setting in which a wide neural network becomes equivalent to kernel regression, a classical and analytically tractable statistical model.

[Setup] Consider a fully connected L-layer multilayer perceptron (MLP) with input dimension d, elementwise nonlinearity σ, hidden-layer widths $n_1, \ldots, n_{L-1}$, and a scalar output. For an input $\mathbf{x} \in \mathbb{R}^d$, the activation $h_\ell(\mathbf{x})$ at layer ℓ and the network output $f(\mathbf{x})$ are defined recursively as:

$$h_0(\mathbf{x}) = \mathbf{x}, \qquad h_{\ell+1}(\mathbf{x}) = \sigma\left(\frac{1}{\sqrt{n_\ell}} W^{(\ell)} h_\ell(\mathbf{x})\right) \quad (\ell = 0, \ldots, L-1), \qquad f(\mathbf{x}) = \frac{1}{\sqrt{n_L}} w^\top h_L(\mathbf{x}).$$

We train the MLP with dataset $\mathscr{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ sampled independently from the data distribution, with $y_i = f_T(\mathbf{x}_i)$. Denote $\mathbf{X} = \{\mathbf{x}_1, \ldots, \mathbf{x}_N\}$ as the training inputs, $\mathbf{y} = (y_1, \ldots, y_N)^\top$ as the corresponding target vector, and $\theta = \mathbf{vec}\{W^{(0)}, \ldots, W^{(L-1)}, w\} \in \mathbb{R}^{P \times 1}$ as the vectorization of all trainable weights. The training loss $\mathcal{L}(\theta)$ is defined as the empirical squared loss:

$$\mathcal{L}(\theta) = \frac{1}{2}\|f_\theta(\mathbf{X}) - \mathbf{y}\|_2^2.$$

In continuous time, the parameters update according to gradient descent with learning rate η:

$$\dot{\theta}_t = -\eta \nabla_\theta \mathcal{L}(\theta_t).$$

[Training Dynamics] Our goal is to find the analytical expression for the learned function f after training. To track how the network output function f evolves during training, we first evaluate it on the training inputs $\mathbf{X}$. Apply the chain rule to transform gradient flow in parameter space into dynamics in function space:

$$\dot{f}_t(\mathbf{X}) \overset{\text{chain rule}}{=} \underbrace{\nabla_\theta f_t(\mathbf{X})}_{\mathbb{R}^{N\times P}} \underbrace{\dot{\theta}_t}_{\mathbb{R}^{P\times 1}} \overset{\text{gradient flow}}{=} -\eta \underbrace{\nabla_\theta f_t(\mathbf{X}) \nabla_\theta f_t(\mathbf{X})^\top}_{\text{NTK} \in \mathbb{R}^{N\times N}} \underbrace{\nabla_{f_t(\mathbf{X})}\mathcal{L}}_{\mathbb{R}^{N\times 1}} \overset{\text{define NTK}}{=} -\eta\, K_t(\mathbf{X}, \mathbf{X}) \nabla_{f_t(\mathbf{X})}\mathcal{L}.$$

[NTK] The key object that appears in the expansion above is the Gram matrix of the Jacobian, which can be interpreted as a kernel function evaluated on pairs of training data. A kernel is a function that takes two inputs and returns a scalar similarity measure. It can be written as an inner product between feature vectors ϕ: $K(\mathbf{x}, \mathbf{x}') = \langle \phi(\mathbf{x}), \phi(\mathbf{x}') \rangle$. In our case, the feature map is the Jacobian $\phi(\mathbf{x}) = \nabla_\theta f_t(\mathbf{x})$, and the resulting kernel was first defined as the Neural Tangent Kernel (NTK) by Jacot et al. (2018):

$$K_t(\mathbf{x}, \mathbf{x}') := \underbrace{\nabla_\theta f_t(\mathbf{x})^\top}_{\mathbb{R}^{1\times P}} \underbrace{\nabla_\theta f_t(\mathbf{x}')}_{\mathbb{R}^{P\times 1}} = \sum_{p=1}^{P} \frac{\partial f_t(\mathbf{x})}{\partial \theta_p} \frac{\partial f_t(\mathbf{x}')}{\partial \theta_p}.$$

If we let all hidden-layer widths $n_\ell \to \infty$, initialize the weights as $W_{ij}^{(\ell)} \sim \mathcal{N}(0,1)$, and use a small learning rate, the NTK converges to a deterministic kernel at initialization and remains constant throughout training (Jacot et al., 2018 ; Lee et al., 2019). This setting, in which the NTK remains stationary during training, is often referred to as the NTK regime, or lazy regime. The $1/\sqrt{n_\ell}$ scaling in the network preactivations is key to lazy learning: under this parameterization, the kernel's deviation from its initialization is bounded as $\sup_{t\geq 0}\|K_t - K_0\|_F = \mathcal{O}(1/\sqrt{n})$. This deviation vanishes as $n \to \infty$, so the NTK remains fixed at its initial value throughout training:

$$K_t \overset{n\to\infty}{\longrightarrow} K_0.$$

[Solving the Dynamics with Stationary NTK] With the fixed NTK, the function-space dynamics therefore reduce to linear ordinary differential equations that can be solved analytically. Denote $K = K_0(\mathbf{X}, \mathbf{X})$ as the stationary NTK Gram matrix on the training inputs. Since the weights are initialized from a mean-zero Gaussian distribution, the network function f at initialization converges to a mean-zero Gaussian process. For simplicity, we study the centered dynamics and take $f_0 = 0$.

The training dynamics are solved as:

$$\dot{f}_t(\mathbf{X}) = -\eta K \nabla_{f_t(\mathbf{X})}\mathcal{L} \overset{\nabla_{f_t(\mathbf{X})}\mathcal{L} = f_t(\mathbf{X}) - \mathbf{y}}{\Longrightarrow} \dot{f}_t(\mathbf{X}) = -\eta K\big(f_t(\mathbf{X}) - \mathbf{y}\big) \Longrightarrow f_t(\mathbf{X}) = \big(I - e^{-\eta K t}\big)\mathbf{y}.$$

The dynamics for a test input $\mathbf{x}_{\text{test}}$ are solved as:

$$\dot{f}_t(\mathbf{x}_{\text{test}}) = -\eta k_{\text{test}}^\top \big(f_t(\mathbf{X}) - \mathbf{y}\big) \Longrightarrow f_t(\mathbf{x}_{\text{test}}) = \eta k_{\text{test}}^\top \int_0^t e^{-\eta K s}\, ds\, \mathbf{y} \overset{\eta\int_0^t e^{-\eta K s} ds = K^{-1}(I - e^{-\eta K t})}{\Longrightarrow} f_t(\mathbf{x}_{\text{test}}) = k_{\text{test}}^\top K^{-1}\big(I - e^{-\eta K t}\big)\mathbf{y},$$

where the test-kernel vector $k_{\text{test}} = K_0(\mathbf{X}, \mathbf{x}_{\text{test}}) = \big[K_0(\mathbf{x}_1, \mathbf{x}_{\text{test}}), \ldots, K_0(\mathbf{x}_N, \mathbf{x}_{\text{test}})\big]^\top \in \mathbb{R}^{N\times 1}$.

[Learned Predictor = Kernel Regression Solution] If K is positive definite, then $e^{-\eta K t} \to \mathbf{0}$ as $t \to \infty$. The converged predictor therefore satisfies

$$f_\infty(\mathbf{X}) = \mathbf{y}, \qquad f_\infty(\mathbf{x}_{\text{test}}) = k_{\text{test}}^\top K^{-1}\mathbf{y}.$$

The prediction at the test point exactly matches the ridgeless kernel regression solution, or equivalently the zero-regularization limit of kernel ridge regression (KRR), a well-studied nonparametric statistical method in function space. This equivalence allows us to approximate the behavior of a wide MLP using KRR and analyze its generalization through the established theory of kernel methods.